

FRAUD INTELLIGENCE PLATFORM

Behavioral Fraud Detection and Explainable Real-Time Transaction Analysis

Feature Guide and Test Examples | All example results were recorded from the running application on 7 October 2026

Part 1. What is this project?

The Fraud Intelligence Platform is a web application that checks a financial transaction against **that customer's own previous behavior**. It does not ask "is this a big payment?" but "is this unusual **for this customer**?". For every transaction it produces:

Output	Meaning
Risk Score (0 to 100)	How risky the customer's recent history looked before this transaction
Fraud Score (0 to 100)	How suspicious this transaction looks to the model
Alert Level	Low, Medium, High or Critical, taken from the Fraud Score
SHAP explanations	Which factors pushed the Fraud Score up, and by how much
Behavioral Similarity	How closely the transaction matches the customer's normal pattern
Customer context	The customer's usual device, home city and history size
Fraud-ring relationships	Other customers linked to this one through a shared device (Fraud Rings tab)

Fraud Score is a model score from 0 to 100. It is NOT a calibrated probability.

A Fraud Score of 99.9 does **not** mean a 99.9% probability of fraud. It means the model ranks the transaction as highly suspicious.

How the examples in this guide were produced

Model	Production model set (production - f42f2a4f6d1d), the default of the application
Database	A new, empty test database (guide_test.db) on a test copy of the project. Your databases were not used or changed.
Method	Every example was sent to the running backend through the same scoring endpoint that Live Scan uses, and the returned values were copied into this guide. The screenshot in Part 6 was made by typing into the Live Scan screen.
Time of the scan	Live Scan always uses the current clock time, and the hour of the day affects the score. To match a daytime presentation, the scan time of each example was set to a daytime hour: 11:15 for the eight detailed examples, and each hour from 9:00 to 18:00 for the cheat sheet.
Customers	A different customer for every example (47 customers in total), because each scan is saved and changes that customer's later scores. None of them had been scanned in your databases when this guide was made.

Before you rely on any example

Results depend on **the hour of the day**, on **the customer's saved history** and on the loaded model. Use the customer listed for your hour (Part 13), scan each customer only once, and expect small differences in the SHAP values, which vary slightly from run to run.

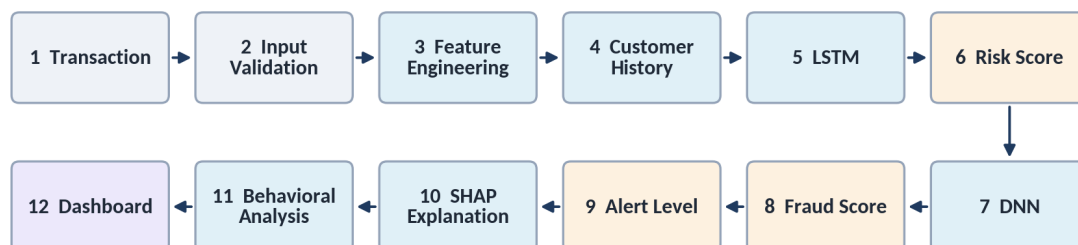
Part 2. Live Scan: every input field

Input	What it means	Example	What happens if you change it?
Customer (customer_id)	Identifies the customer. The system loads this customer's previous transactions and compares the new one with them. Every customer has a different baseline.	CUST_0219	Everything is judged against a different history, so the same inputs can give a different result.

Input	What it means	Example	What happens if you change it?
Amount (Rs.) (amount)	Transaction amount in rupees. It is compared with the customer's own average, not with a fixed limit. Rs. 4,000 is normal for a customer who averages Rs. 3,993 and ten times the average for one who averages Rs. 380.	1,500 5,000 20,000 75,000	Lowers Behavioral Similarity. In our tests the production model did not raise the Fraud Score for amount alone (Example 5); amount matters together with other signals.
Merchant Category (merchant_category)	The type of merchant. The form offers ten: grocery, electronics, travel, dining, utilities, entertainment, fashion, healthcare, fuel, online_retail. Each customer normally uses three to five of them.	grocery	A category the customer does not normally use raised the score to Medium on its own (Example 2).
Device ID (device_id)	The device used. Leave it blank to use the customer's usual device. Any ID the customer has never used counts as a new device.	DEV_0219_A DEV_UNKNOWN N_5001	A new device alone moved the score only a little (Example 6). With a failed login and a higher amount it reached High (Example 3).
Location (location)	The city. Leave it blank to use the home city. A city the customer never used is a new location. A city outside the ten home cities of the data (Hyderabad, Mumbai, Delhi, Bangalore, Chennai, Kolkata, Pune, Ahmedabad, Jaipur, Lucknow) also counts as foreign.	Pune Mumbai Singapore	A new city on the list: small effect (Example 7b). A foreign city: Critical on its own (Example 7).
Failed Logins (24h) (failed_logins_24h)	Number of failed login attempts in the previous 24 hours. Zero is normal.	0 2 5	On their own, even 5 failed logins left the score Low (Example 8). Together with a new device they pushed it up (Example 3).

The two **Quick scenario** buttons fill fixed values that are the same for every customer (Typical Purchase: Rs. 1,500 grocery), so they are not typical for everyone. For a planned demonstration, type the values from Part 13.

Part 3. How the system generates the result



Step	What happens
Input validation	Checks the fields: a customer, an amount above 0, a category, a device, a location, failed logins of 0 or more.
Customer history	Loads the customer's past transactions in time order (seed data plus every earlier scan).
9 behavioral features	Nine numbers that compare the transaction with that history: amount_zscore, amount_pct_of_avg, hour_is_unusual (1 if under 5% of the customer's past transactions were in this hour), is_new_device, is_new_location, is_foreign_location, category_is_unusual, failed_logins_24h, txn_velocity_1h (transactions in the last hour, so a second scan soon after the first already differs).
LSTM → Risk Score	The LSTM reads the customer's previous 10 transactions and gives the Risk Score.
DNN → Fraud Score	The DNN takes the 9 features of the current transaction + the Risk Score (10 inputs) and gives the Fraud Score.
Alert, SHAP, Similarity	Alert Level from fixed bands (Part 9); SHAP lists up to four features that pushed the score up; Similarity measures the distance from the customer's own averages.
Dashboard / PDF	Shows the result, saves it to the database, and offers a PDF report.

Part 4. Test examples with real results

All eight examples: production model, scan time 11:15, one fresh customer each. "Customer's normal pattern" comes from that customer's seed history.

Example 1: Normal transaction (Low Risk)

What was entered	Customer's normal pattern	What the system returned
Customer: CUST_0219 Amount: Rs. 2,590 Merchant: fashion Device: DEV_0219_A (usual) Location: Pune (home) Failed logins: 0	Usual device: DEV_0219_A Home city: Pune Average amount: Rs. 2,593 Usual categories: fashion, grocery, online_retail, dining History: 173 transactions	Alert Level: Low Risk Fraud Score: 1.67 / 100 Risk Score: 22.88 / 100 Similarity: 95.25% Deviation: 4.75% Transaction ID: TXN_19769AD2E4
SHAP reasons None listed	Why this result? Everything matches the customer: the amount equals the average, fashion is the most used category, the device and city are the usual ones, there are no failed logins, and 11:00 is one of the customer's active hours. Similarity is 95% and SHAP has nothing to report.	

Example 2: Normal but different (Medium Risk)

What was entered	Customer's normal pattern	What the system returned
Customer: CUST_0240 Amount: Rs. 2,180 Merchant: dining Device: DEV_0240_A (usual) Location: Pune (home) Failed logins: 0	Usual device: DEV_0240_A Home city: Pune Average amount: Rs. 2,181 Usual categories: fuel, electronics, fashion History: 196 transactions	Alert Level: Medium Risk Fraud Score: 27.14 / 100 Risk Score: 11.35 / 100 Similarity: 70.65% Deviation: 29.35% Transaction ID: TXN_B4B38D2F0B
SHAP reasons Unusual Merchant Category +0.211 High Transaction Velocity +0.040 Elevated Behavioral Risk Score +0.000	Why this result? Only one thing is different: this customer has never bought in the dining category. Amount, device, city and logins are normal. SHAP names the unusual category as the reason, and similarity falls to 71%. We used the category here because changing only the amount did not move the Fraud Score (see Example 5).	

Example 3: Suspicious (High Risk)

What was entered	Customer's normal pattern	What the system returned
Customer: CUST_0232 Amount: Rs. 1,390 Merchant: entertainment Device: DEV_UNKNOWN_6047 (new) Location: Mumbai (home) Failed logins: 1	Usual device: DEV_0232_A Home city: Mumbai Average amount: Rs. 693 Usual categories: entertainment, fuel, grocery, fashion, electronics History: 272 transactions	Alert Level: High Risk Fraud Score: 69.38 / 100 Risk Score: 11.49 / 100 Similarity: 31.83% Deviation: 68.17% Transaction ID: TXN_647EC07615
SHAP reasons New Device +0.626 Multiple Failed Logins +0.205	Why this result? Three things changed together: a device the customer never used, one failed login, and twice the usual amount. SHAP gives most of the weight to the new device (+0.626) and the rest to the failed login (+0.205). City and category are normal, which keeps it below Critical. Adding a foreign city or an unusual category to this made it Critical in our trials, so those were left out.	

Example 4: Very suspicious (Critical Risk)

What was entered	Customer's normal pattern	What the system returned
Customer: CUST_0254 Amount: Rs. 85,000 Merchant: travel Device: DEV_UNKNOWN_7429 (new) Location: Singapore (not home) Failed logins: 4	Usual device: DEV_0254_A Home city: Bangalore Average amount: Rs. 3,704 Usual categories: utilities, electronics, grocery, online_retail History: 224 transactions	Alert Level: Critical Risk Fraud Score: 99.61 / 100 Risk Score: 13.46 / 100 Similarity: 0.00% Deviation: 100.00% Transaction ID: TXN_A7F60DA8AF
SHAP reasons Unusual Merchant Category +0.283 Foreign Location +0.222 New Device +0.217 Unusual Transaction Amount +0.174	Why this result? Everything is abnormal at once: Rs. 85,000 against an average of Rs. 3,704, a category never used, a new device, a foreign city and four failed logins. Similarity is 0%. Notice the Risk Score is only 13.46: the customer's history was calm, and it is this transaction that is the problem.	

Example 5: High amount but normal behavior (Low Risk)

What was entered	Customer's normal pattern	What the system returned
Customer: CUST_0233 Amount: Rs. 3,990 Merchant: dining Device: DEV_0233_A (usual) Location: Delhi (home) Failed logins: 0	Usual device: DEV_0233_A Home city: Delhi Average amount: Rs. 3,993 Usual categories: dining, fashion, online_retail, electronics History: 184 transactions	Alert Level: Low Risk Fraud Score: 3.28 / 100 Risk Score: 4.83 / 100 Similarity: 95.05% Deviation: 4.95% Transaction ID: TXN_C17D5ED507
SHAP reasons None listed	Why this result? Large amount does not mean fraud. Rs. 3,990 is one of the highest average spends in the data, but for this customer it is simply the average, on the usual device, in the home city, in the favourite category. Similarity is 95% and the result is Low.	

Example 5b: The same Rs. 3,990 for a small spender

What was entered	Customer's normal pattern	What the system returned
Customer: CUST_0230 Amount: Rs. 3,990 Merchant: online_retail Device: DEV_0230_A (usual) Location: Kolkata (home) Failed logins: 0	Usual device: DEV_0230_A Home city: Kolkata Average amount: Rs. 380 Usual categories: online_retail, electronics, travel, grocery History: 268 transactions	Alert Level: Low Risk Fraud Score: 0.87 / 100 Risk Score: 17.33 / 100 Similarity: 44.30% Deviation: 55.70% Transaction ID: TXN_1A063672DB
SHAP reasons None listed	Why this result? Same amount, but this customer averages Rs. 380, so it is more than ten times normal. Behavioral Similarity drops from 95% to 44% , which shows the behavioral comparison at work. The Fraud Score, however, stays Low: the production model does not react to amount on its own. This is honest behavior of the current model and a good point to mention if asked about limitations.	

Example 6: New device only

What was entered	Customer's normal pattern	What the system returned
Customer: CUST_0260 Amount: Rs. 1,380 Merchant: grocery Device: DEV_UNKNOWN_7714 (new) Location: Chennai (home) Failed logins: 0	Usual device: DEV_0260_A Home city: Chennai Average amount: Rs. 1,384 Usual categories: grocery, online_retail, utilities, dining History: 257 transactions	Alert Level: Low Risk Fraud Score: 5.42 / 100 Risk Score: 7.55 / 100 Similarity: 62.18% Deviation: 37.82% Transaction ID: TXN_BB7AE94D84
SHAP reasons New Device +0.034 High Transaction Velocity +0.009 Unusual Transaction Amount +0.002	Why this result? Only is_new_device changed. SHAP lists New Device, but with a small contribution (+0.034), and the Fraud Score stays Low at 5.42. Similarity falls to 62%. A new phone alone is common for genuine customers; the model starts to react when it is combined with other signals, as in Example 3.	

Example 7: Foreign location only

What was entered	Customer's normal pattern	What the system returned
Customer: CUST_0290 Amount: Rs. 1,430 Merchant: electronics Device: DEV_0290_A (usual) Location: Singapore (not home) Failed logins: 0	Usual device: DEV_0290_A Home city: Hyderabad Average amount: Rs. 1,430 Usual categories: electronics, utilities, healthcare History: 193 transactions	Alert Level: Critical Risk Fraud Score: 84.90 / 100 Risk Score: 25.50 / 100 Similarity: 54.32% Deviation: 45.68% Transaction ID: TXN_C7B5154F69
SHAP reasons Foreign Location +0.455 New Location +0.227 High Transaction Velocity +0.040 Amount Far Above Average +0.012	Why this result? Only the city changed, to one outside the ten home cities. Both Foreign Location (+0.455) and New Location (+0.227) appear, and the transaction is Critical on location alone. For the production model this is the strongest single signal.	

Example 7b: New Indian city only

What was entered	Customer's normal pattern	What the system returned
Customer: CUST_0382 Amount: Rs. 1,340 Merchant: entertainment Device: DEV_0382_A (usual) Location: Mumbai (not home) Failed logins: 0	Usual device: DEV_0382_A Home city: Hyderabad Average amount: Rs. 1,342 Usual categories: entertainment, online_retail, travel, healthcare, utilities History: 213 transactions	Alert Level: Low Risk Fraud Score: 13.36 / 100 Risk Score: 10.77 / 100 Similarity: 54.75% Deviation: 45.25% Transaction ID: TXN_33F591BE38
SHAP reasons New Location +0.061 High Transaction Velocity +0.008 Amount Far Above Average +0.007 Multiple Failed Logins +0.004	Why this result? The city is new for this customer but is one of the ten home cities, so it is a new location and not a foreign one. The effect is small: Fraud Score 13.36, Low. Compare with Example 7 to explain the difference between is_new_location and is_foreign_location.	

Example 8: Failed login attack

What was entered	Customer's normal pattern	What the system returned
Customer: CUST_0405 Amount: Rs. 1,680 Merchant: electronics Device: DEV_0405_A (usual) Location: Chennai (home) Failed logins: 5	Usual device: DEV_0405_A Home city: Chennai Average amount: Rs. 1,676 Usual categories: electronics, healthcare, grocery, utilities History: 156 transactions	Alert Level: Low Risk Fraud Score: 4.64 / 100 Risk Score: 24.74 / 100 Similarity: 51.65% Deviation: 48.35% Transaction ID: TXN_B36F8DDF58
SHAP reasons None listed	Why this result? Five failed logins, everything else normal. Similarity falls to 52% because the customer normally has none, but the Fraud Score stays Low (4.64) and SHAP lists nothing. The production model does not treat failed logins alone as fraud; they add weight when a new device is also present (Example 3: +0.205).	

Part 5. Master example table

Example	Customer	Amount	Merchant	Device	Location	Logins	Risk	Fraud	Similarity	Alert
Normal	CUST_0219	2,590	fashion	DEV_0219_A	Pune	0	22.88	1.67	95.25%	Low
Medium	CUST_0240	2,180	dining	DEV_0240_A	Pune	0	11.35	27.14	70.65%	Medium
High	CUST_0232	1,390	entertainment	DEV_UNKNOWN_6047	Mumbai	1	11.49	69.38	31.83%	High
Critical	CUST_0254	85,000	travel	DEV_UNKNOWN_7429	Singapore	4	13.46	99.61	0.00%	Critical
Large but normal	CUST_0233	3,990	dining	DEV_0233_A	Delhi	0	4.83	3.28	95.05%	Low
Same amount, small spender	CUST_0230	3,990	online_retail	DEV_0230_A	Kolkata	0	17.33	0.87	44.30%	Low
New device	CUST_0260	1,380	grocery	DEV_UNKNOWN_7714	Chennai	0	7.55	5.42	62.18%	Low
Foreign location	CUST_0290	1,430	electronics	DEV_0290_A	Singapore	0	25.50	84.90	54.32%	Critical
New Indian city	CUST_0382	1,340	entertainment	DEV_0382_A	Mumbai	0	10.77	13.36	54.75%	Low
Failed login	CUST_0405	1,680	electronics	DEV_0405_A	Chennai	5	24.74	4.64	51.65%	Low

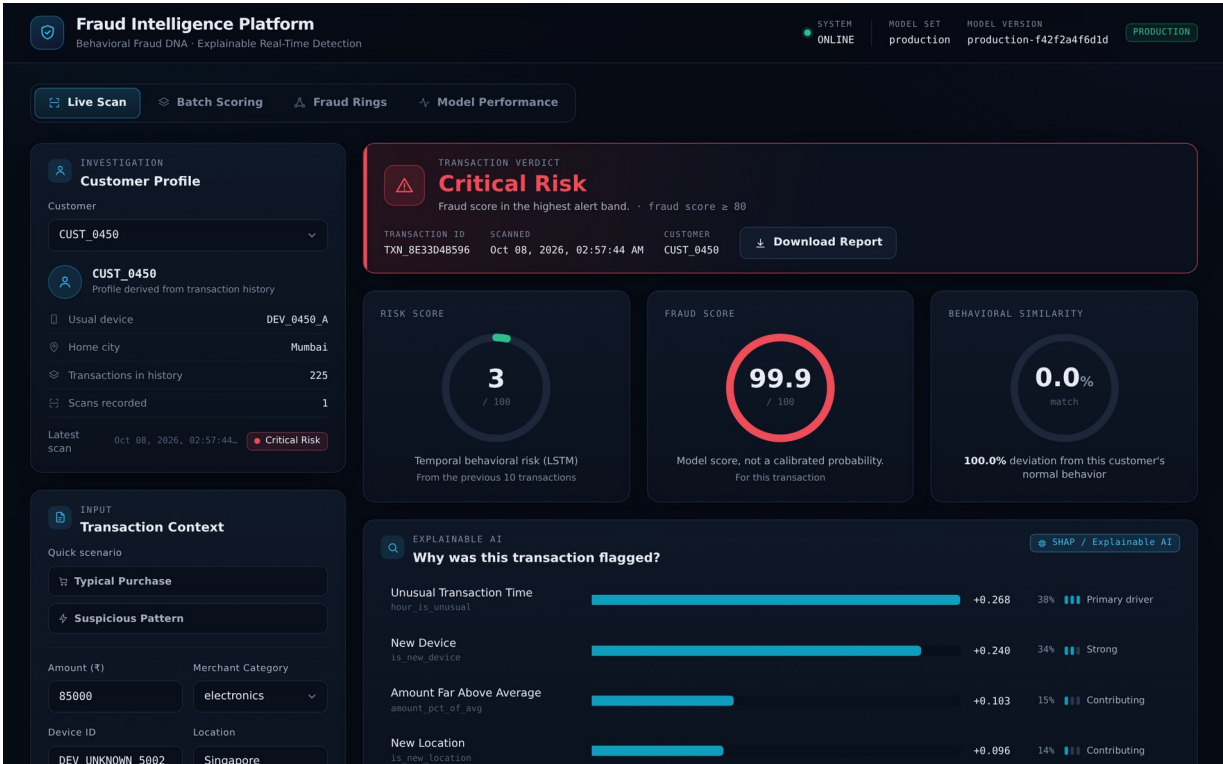
Logins = failed logins in 24 hours. Risk = Risk Score. Fraud = Fraud Score.

What these tests show about the production model

- **Strong on its own:** a foreign location (Critical). An hour unusual for the customer also raises the score strongly (see the screenshot in Part 6).
- **Moderate on its own:** an unusual merchant category (Medium).
- **Weak on its own:** a new device, a new Indian city, failed logins, and a high amount (all Low).
- **Combinations escalate:** new device + failed login + double amount gave High; everything together gave Critical.
- **Behavioral Similarity reacts to every change,** even when the Fraud Score does not.

Part 6. SHAP explanation guide

SHAP answers one question: **"Why did the model give this transaction this score?"** It splits the score into contributions from the individual features. The Live Scan screen shows the features that pushed the Fraud Score up, largest first.



Real screenshot: CUST_0450, Rs. 85,000, electronics, DEV_UNKNOWN_5002, Singapore, 4 failed logins, typed into Live Scan. Result: Critical Risk, Fraud Score 99.9, Risk Score 3.48, Similarity 0.00%.

What the screen shows	How to read it
Unusual Transaction Time +0.268 New Device +0.240 Amount Far Above Average +0.103 New Location +0.096	Each line is one feature. A positive value means the feature pushed the Fraud Score upward . The bar length and the percentage show its share of the listed reasons. "Primary driver", "Strong" and "Contributing" are labels added by the screen from these values.

In this screenshot the first reason is Unusual Transaction Time: the scan was made at an hour when this customer is normally inactive. That is the time-of-day effect described in Part 13.

Remember

SHAP explains the **model's prediction**. It does **not** prove that fraud occurred. Very small values (below about 0.001) are noise from the sampling SHAP uses and can be ignored.

Part 7. Behavioral Similarity

Similarity: how similar is the transaction to the customer's historical behavior? The system measures how far each of the nine features is from the customer's own average and converts the result to a percentage.	Deviation = 100% – Similarity. Example 1: similarity 95.25%, deviation 4.75%. Example 4: similarity 0.00%, deviation 100.00%.
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Similarity	Simple interpretation	Seen in our examples
High	Very similar to normal behavior	95% for the normal transactions (Examples 1 and 5)
Medium	Somewhat different	52% to 71% when one thing changed (Examples 2, 6, 7, 7b, 8)
Low	Strongly different	0% to 44% (Examples 3, 4, 5b)

The system does **not** define fixed cut-offs for high, medium and low similarity; the ranges above are only what we observed. Similarity is a separate indicator. It is not an input to the models, which is why it can be low while the Fraud Score is also low (Example 5b). The screen also puts the usual device and home city next to the ones in the transaction and marks them Match or Different.

Part 8. Risk Score vs Fraud Score

Risk Score	Fraud Score
Based on previous behavior	Based on the current transaction
Produced by the LSTM	Produced by the DNN
Looks at the previous 10 transactions	Uses the 9 current behavioral features + the Risk Score
A temporal risk signal from the recent sequence	Describes transaction-level suspicion
Does not change when you change the form values	Changes with the amount, device, location, category and logins you enter

Example. In Example 4 the Risk Score is 13.46 and the Fraud Score is 99.61. The Risk Score is low because the customer's last ten transactions were ordinary. The Fraud Score is high because this transaction is completely unlike them. A fraudster's first transaction on a healthy account looks exactly like this: calm history, suspicious transaction. After several suspicious transactions are saved for a customer, the Risk Score for their next transaction rises as well.

Part 9. Alert Levels

Alert Level	Rule in the code	What it means in practice
Low Risk	Fraud Score below 25	No alert. The transaction looks normal for this customer.
Medium Risk	Fraud Score from 25 to below 50	Something is unusual. Worth a look if time allows.
High Risk	Fraud Score from 50 to below 80	Several signals together. Should be reviewed.
Critical Risk	Fraud Score 80 and above	Strongest alerts. Review first.

Alert level is a triage classification, not proof of fraud. The bands are fixed values in the code (alert_level_from_probability); they were not tuned from an evaluation of the model.

Part 10. Batch Scoring

Steps 1. Prepare a CSV file with a header row. 2. Open Batch Scoring and choose or drop the file. 3. The screen checks the header (format check). 4. Click Upload & Score. 5. Wait: 30 rows took about 20 seconds on your laptop. 6. Review the alert distribution. 7. Filter by Critical, High, Medium or Low. 8. Click Download results for a CSV of the scores.	Required columns • customer_id • amount • merchant_category Optional columns • device_id (blank = usual device) • location (blank = home city) • failed_logins_24h (blank = 0) If any row is invalid, nothing is scored and the bad rows are listed.
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Sample CSV (customers that exist in the seed data; these rows show the format and were not scored for this guide):

customer_id	amount	merchant_category	device_id	location	failed_logins_24h
CUST_0412	2400	grocery			0
CUST_0415	1850	dining			0
CUST_0420	6200	fashion			1
CUST_0423	48000	electronics	DEV_UNKNOWN_5101	Dubai	3
CUST_0427	900	fuel		Jaipur	0
CUST_0431	72000	travel	DEV_UNKNOWN_5102	Singapore	5

Ready-made files are in test_data\batch_scoring\ (15, 15 and 30 rows). Every upload is saved, so upload a file once, or use a separate database for practice.

Part 11. Fraud Rings

Term	Meaning
Customer / Device	The two kinds of points in the graph: round = customer, square = device.
Shared device	A device that appears in the history of two or more customers.
Connected customers	Customers who used the same shared device.
Cluster	A group of customers and devices that are all connected, directly or through each other.
Investigation summary	For the selected cluster: customers linked, shared devices, linked transactions, and a severity label based only on the size of the cluster.

Example Customer A → Device X Customer B → Device X Customer C → Device X Therefore A, B and C are connected through Device X.	In our data the page shows 20 shared devices, 29 affected customers and 9 clusters. The largest cluster links 7 customers through 6 devices. This is a suspicious relationship, NOT proof of fraud. Family members can share a phone. It is a rule over device IDs, not a machine-learning model. If you use the same new device ID on two customers in Live Scan, a new ring appears here.
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Part 12. Model Performance

On the screen	What does it tell me?
Recall	Of all fraud transactions, what share did the model flag? Higher is better.
Precision	Of everything the model flagged, what share was really fraud? Higher means fewer false alarms.
PR-AUC	One number for the balance between catching fraud and avoiding false alarms. The right summary when fraud is rare.
ROC-AUC	How well the model ranks fraud above genuine transactions. 0.5 is random, 1.0 is perfect.
Fraud caught	How many of the fraud transactions in the test data were flagged, as a count.
Alert rate	Alerts per 1,000 transactions: the workload for investigators. Alerts on genuine transactions are the "legitimate alerts".
Confusion matrix	Four counts: genuine correctly passed, genuine wrongly flagged (false positive), fraud missed (false negative), fraud caught.
Model information	Which model is loaded, its version, its data and features, and what the scores mean.

The page shows 1.000 for the production classifier. That is measured on the first, easy dataset (v1), where a two-line rule also scores 1.000. On the harder final hold-out the production model's recall is 27.9%, and the Seed-14 candidate reaches 55.8% but is not deployed.

Part 13. Presentation demo cheat sheet

Find the hour of your presentation, then type the row for the level you want to show. Each row was run once on a clean database at that hour and gave the result shown. Use each customer once.

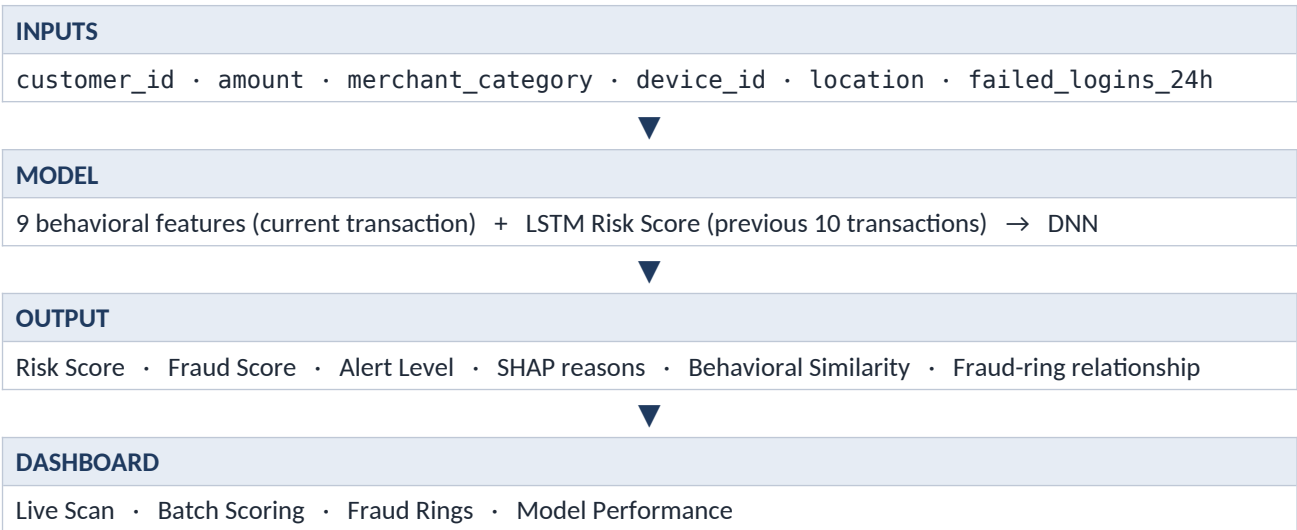
Hour	Show	Customer	Amount	Merchant	Device	Location	Logins	Actual result
9:00	Normal	CUST_0127	2,930	electronics	usual	home	0	Low (2.20)
9:00	Medium	CUST_0246	2,240	dining	usual	home	0	Medium (29.52)
9:00	High	CUST_0270	3,920	electronics	DEV_UNKNOWN_8613	home	1	High (63.26)
9:00	Critical	CUST_0251	85,000	electronics	DEV_UNKNOWN_6375	Singapore	4	Critical (99.54)
10:00	Normal	CUST_0144	2,770	dining	usual	home	0	Low (1.60)
10:00	Medium	CUST_0189	3,800	entertainment	usual	home	0	Medium (31.78)
10:00	High	CUST_0215	890	dining	DEV_UNKNOWN_6788	home	2	High (63.63)
10:00	Critical	CUST_0259	85,000	fashion	DEV_UNKNOWN_8279	Singapore	4	Critical (99.57)
11:00	Normal	CUST_0219	2,590	fashion	usual	home	0	Low (1.67)
11:00	Medium	CUST_0240	2,180	dining	usual	home	0	Medium (27.14)
11:00	High	CUST_0232	1,390	entertainment	DEV_UNKNOWN_6047	home	1	High (69.38)
11:00	Critical	CUST_0254	85,000	travel	DEV_UNKNOWN_7429	Singapore	4	Critical (99.61)
12:00	Normal	CUST_0191	2,720	utilities	usual	home	0	Low (1.44)
12:00	Medium	CUST_0334	440	dining	usual	home	0	Medium (27.71)
12:00	High	CUST_0250	2,160	online_retail	DEV_UNKNOWN_7834	home	2	High (65.75)
12:00	Critical	CUST_0365	85,000	electronics	DEV_UNKNOWN_5001	Singapore	4	Critical (99.62)
13:00	Normal	CUST_0266	400	fashion	usual	home	0	Low (3.40)
13:00	Medium	CUST_0360	3,760	entertainment	usual	home	0	Medium (34.91)
13:00	High	CUST_0324	2,910	fuel	DEV_UNKNOWN_2388	home	2	High (56.47)
13:00	Critical	CUST_0305	85,000	electronics	DEV_UNKNOWN_4100	Singapore	4	Critical (99.62)
14:00	Normal	CUST_0293	420	online_retail	usual	home	0	Low (3.79)
14:00	Medium	CUST_0325	1,660	dining	usual	home	0	Medium (29.38)
14:00	High	CUST_0347	720	fuel	DEV_UNKNOWN_3995	home	1	High (57.33)
14:00	Critical	CUST_0371	85,000	electronics	DEV_UNKNOWN_8114	Singapore	4	Critical (99.67)
15:00	Normal	CUST_0291	3,710	dining	usual	home	0	Low (1.46)
15:00	Medium	CUST_0400	1,110	grocery	usual	home	0	Medium (32.18)
15:00	High	CUST_0438	4,840	electronics	DEV_UNKNOWN_8387	home	3	High (66.83)
15:00	Critical	CUST_0472	85,000	electronics	DEV_UNKNOWN_2036	Singapore	4	Critical (99.59)
16:00	Normal	CUST_0273	2,600	utilities	usual	home	0	Low (1.42)
16:00	Medium	CUST_0356	390	dining	usual	home	0	Medium (29.81)
16:00	High	CUST_0352	1,780	grocery	DEV_UNKNOWN_5498	home	3	High (59.96)
16:00	Critical	CUST_0317	85,000	electronics	DEV_UNKNOWN_7747	Singapore	4	Critical (99.63)
17:00	Normal	CUST_0299	1,790	dining	usual	home	0	Low (1.68)
17:00	Medium	CUST_0367	2,120	fuel	usual	home	0	Medium (27.26)
17:00	High	CUST_0349	1,500	dining	DEV_UNKNOWN_2919	home	1	High (73.69)
17:00	Critical	CUST_0374	85,000	electronics	DEV_UNKNOWN_7578	Singapore	4	Critical (99.75)
18:00	Normal	CUST_0322	3,180	grocery	usual	home	0	Low (3.76)
18:00	Medium	CUST_0474	1,130	dining	usual	home	0	Medium (36.63)
18:00	High	CUST_0403	1,700	entertainment	DEV_UNKNOWN_2833	home	2	High (65.46)
18:00	Critical	CUST_0409	85,000	travel	DEV_UNKNOWN_5275	Singapore	4	Critical (99.62)

"usual" and "home" mean: leave Device ID and Location **blank**. The number in brackets is the Fraud Score. The 11:00 rows are Examples 1 to 4. Outside 9:00 to 18:59 nothing was checked.

Part 14. Viva questions

#	Question	Short answer
1	What problem are you solving?	Detecting suspicious financial transactions, explaining why they are suspicious, and finding accounts that are linked.
2	Why behavioral fraud detection?	Normal is different for every customer. A fixed rule either misses fraud or flags genuine customers.
3	Why LSTM?	It reads sequences. It models the last 10 transactions to capture temporal behavioral patterns. We do not claim it predicts fraud before it happens.
4	Why DNN?	The final decision is about one transaction described by 10 numbers; a small dense network suits that and is fast.
5	What is Risk Score?	The LSTM's 0 to 100 score for the customer's previous 10 transactions.
6	What is Fraud Score?	The DNN's 0 to 100 score for the current transaction.
7	Is Fraud Score a probability?	No. It is an uncalibrated model score. 99.9 does not mean 99.9% chance of fraud.
8	What is SHAP?	A method that shows how much each feature contributed to one prediction.
9	Why do we need explainability?	An investigator must justify an alert. Reasons can be checked; a bare number cannot.
10	What is behavioral similarity?	How close the transaction is to the customer's own past behavior, as a percentage. Deviation is 100% minus it.
11	What is a fraud ring?	A group of customers connected through shared devices. A lead, not proof.
12	Is fraud-ring detection machine learning?	No. It is a rule: list every device used by two or more customers.
13	What is Batch Scoring?	Uploading a CSV to score many transactions at once, then filtering and exporting the results.
14	Why use synthetic data?	Real banking data is confidential. We generated data with known fraud patterns.
15	Why did you create v2 data?	On v1 even a two-line rule scored 1.000, so it was too easy. v2 has more realistic, overlapping behavior.
16	Why is Seed-14 not deployed?	It passed the final test, but evidence is synthetic, new customers get too many alerts, and its alert margin is thin.
17	What are the limitations?	Synthetic data, uncalibrated scores, weak on new customers, sensitive to time of day, rule-based rings, prototype only.
18	What happens with a new customer?	With fewer than 10 earlier transactions the LSTM is not run and an average Risk Score is used; accuracy is weaker.
19	What technologies did you use?	Python, FastAPI, TensorFlow/Keras, SHAP, SQLite; React, Vite, Tailwind, Recharts; ReportLab for PDFs.
20	What would you improve in the future?	Real data, calibrated scores, a new-customer policy, authentication, streaming and richer graph analysis.

Quick reference



Alert Level	Fraud Score	One input that gave it in our tests
Low	below 25	Everything normal; or only a new device, only failed logins, only a high amount
Medium	25 to below 50	Only an unusual merchant category
High	50 to below 80	New device + failed login + about double the usual amount
Critical	80 and above	A foreign location, alone or with anything else

Five things to remember before presenting

1. **Fraud Score is a model score, not a probability.**
2. **The hour matters.** Pick the customer for your hour from Part 13. The same normal purchase at an unusual hour for that customer can come out Medium or higher.
3. **Scan each customer once.** Every scan is saved and changes that customer's next result. Do not use the same new device ID on two customers unless you want a new ring to appear.
4. **Production is the model on screen.** Seed-14 is a validated candidate that is not deployed.
5. **All data is synthetic.** A ring or an alert is a lead for an investigator, not proof of fraud.